

Tarek Ahmed Radwan

G17, M1 5DB, The Hacienda Apartments

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Education

2021-2022 **Imperial College of London (ICL)**
MSc Applied Machine learning

Example Modules studied:

- | | |
|----------------------------------|-------------------------|
| -Artificial Intelligence | -Computer Vision |
| -Speech Processing | -Deep Learning |
| -Large dimensional Data analysis | -Human-Centred Robotics |

Imperial College MSc thesis (2021 -2022)

IOS application for wheelchair user navigation

- Built an IOS application allowing users to tag wheelchair user barriers
- Barriers are then sent to a PHP server and saved in an SQL database
- Real time object detection of barriers created to detect, localise, and add barriers in run-time
- Custom algorithm implemented to traverse the map and find best path for wheelchair users

2017 – 2021 **The University of Manchester (UoM)**
MEng Mechatronics Engineering

First year result: 82% Second Year Result: 84% Third year Result:80%

Example Modules Studied:

- | | |
|---------------------------|---|
| -Concurrent Systems | - Machine Learning and Optimization |
| -C Programming | - Microcontroller Engineering, I and II |
| -Digital Image Processing | - Data Networking |

University MEng project (2020-2021):

Machine Learning Algorithm for Image Classification and Object Detection

- Me and 7 team members worked a project with Kenny Waste Management to build a waste classifier robot
- Used Python and Facebook's machine learning library (Pytorch) to build an image classification algorithm that can differentiate between different materials e.g., wood, plastic, bricks, etc
- Used YOLOV3 and YOLOV5 for localising and detecting objects before wirelessly sending them to a robot via ROS
- Programmed the robot to pick the detected waste material from the sight
- Coded a website to control the robot remotely

University Third Year project (2019-2020):

Stepper Motor Actuated Robot Arm:

- Used CAD software to design a 3 degree of freedom robot arm
- 3D printed the designed parts and assembled them together
- Created motor driver's PCB for controlling the arm joints
- Equipped the arm with a camera and programmed it to detect a designated object before picking it up

University Second Year project (s) (2018-2019): Embedded systems project:

- Led a group of 5 students to design and build a white line follower robot (buggy)
- Designed the buggy's chassis using CAD software, built its printed circuit boards, schematic diagrams, as well as building a control system (PID controller) to command the buggy to follow the white line

2006 – 2017

Global International School Jeddah, Saudi Arabia

- Cambridge GCSE levels: including Biology, Chemistry, Mathematics, Physics, Accounting, Economics, ICT (A*, A*, A*, A*, A*, A*, A*, A*)
- Cambridge Advanced Subsidiary (AS) levels: including Mathematics, Physics, Chemistry, and Biology (A, A, A, A)
- Cambridge A2 levels: including Physics, Mathematics, and Chemistry (A*, A, A)

Work Experience

Nov 2018 – Feb 2019

Student ambassador, UOM

- Responsible for leading the applicants and parents on a campus tour
- Carried out different activities keeping applicants and parents intrigued as I explained the different buildings available on campus, and described how students used them

Sep 2022 – Present

Cybersecurity Consultant, Deloitte LLP

- Responsible for testing clients web application as well as cloud infrastructure
- Writing reports and building PowerPoint presentations on faults present in client's environment
- Gained Microsoft Azure Fundamentals Certificate and Certified DevSecOps Professional Certificate
- 200+ hours of hands-on web application testing using Burp Suite
- Wrote plenty of Bash and Python scripts for automating tasks
- Built complete CI/CD pipelines to embed security to code on GitLab and Jenkins before deployment
- Very fluent with various programming languages (6 years of experience)
- Currently studying for the BSCP and OSCP certificates

Volunteering and Societies

Sept 2019 – Mar 2020

Manchester Can-Sat Society

- Worked in a group of 8 engineering students to design and construct a can-sized satellite sensors to detect certain weather conditions

Additional Skills

Software/Programming

- Competent in the use of Python (with machine learning libraries such as PyTorch and TensorFlow), Swift (iOS programming), HTML, CSS, JavaScript, C, C++, Arduino IDE, VHDL, Altium Designer, PROLOG, SolidWorks CAD, MATLAB, Simulink, LabView, and MPLABX IDE (machine code), mbed, Keil uVision, Linux terminals, PowerShell, Angular, TypeScript and .NET
- Built various web applications and
- Very familiar with different cloud environments, networking protocols, and cyber frameworks

Languages and Hobbies

Languages

- Languages include Arabic (mother tongue), English (fluent), and French (Basics)

Hobbies

- Football and Bodybuilding